

Neural Network-Based Acoustic Detection of Red Palm Weevil Infestation in Coconut Trees

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Abstract—The red palm weevil (*Rhynchophorus ferrugineus*) poses a serious threat to coconut palms, often causing irreversible damage before visible symptoms appear. Traditional inspection methods are labor-intensive and ineffective for early-stage detection. This study proposes a neural network-based approach for real-time detection of red palm weevil infestations using acoustic signals emitted by the larvae within coconut trees. Sound recordings collected under diverse environmental conditions are preprocessed and used to train a lightweight neural network model capable of identifying infestation patterns based on characteristic audio frequencies. The model's performance is evaluated using key metrics, achieving a testing accuracy of 85.6%, training accuracy of 96.1%, precision of 0.72, recall of 0.80, F1-score of 0.72. Experimental results demonstrate that the proposed method effectively detects red palm weevil activity at an early stage, offering a scalable and low-cost solution for precision agriculture and timely pest management. Furthermore, comparative evaluation with recent ensemble methods such as XGBoost and CatBoost showed that the proposed neural model achieves superior accuracy with lower computational cost, confirming its suitability for real-time field deployment. This acoustic detection technique can be integrated into edge devices for continuous monitoring, enabling early intervention and reducing crop losses.

Index Terms—Red Palm Weevil, Coconut Trees, Acoustic Detection, Neural Network, Pest Monitoring, Precision Agriculture, Sound-Based Detection, Edge Computing.

I. INTRODUCTION

Agriculture functions as the fundamental pillar for numerous nations because it both sustains food supplies and generates work opportunities for billions of individuals around the world. The major threat to crop productivity emerges from pest infestations which lead to extensive losses both in agricultural production and economic value. The primary approach to pest detection involves human visual inspection together with expert analysis as well as field investigations. The methods require extensive human labor and consume

much time while being at risk of human mistakes or inaccurate judgments. The combination of seasonal changes and environmental conditions makes it difficult to detect pests using manual methods which results in operational inefficiencies for large farming areas. AI has been having a real-world impact in agriculture by empowering farmers to fight challenges early and with accuracy. From the identification of plant diseases and tracking the ripeness of fruits to identifying pests like the red palm weevil (RPW), smart technologies are bridging gaps where the human eye may be missing early signs. For RPW, researchers use very tiny audio sensors to pick up sounds generated by larvae inside trees—sounds imperceptible to the human ear. These audio signals are processed with the assistance of AI algorithms like CNN and LSTM that are trained to recognise infestation patterns. Some other systems even report this audio information along with temperature and humidity information to achieve higher accuracy. This pre-detection process gives farmers an early warning of protection for their produce, which means saving time and product. This paper mainly focuses on detecting, the Red Palm Weevil pest. It is an invasive beetle that severely damages coconut and other palm trees worldwide. Originating from South and Southeast Asia, it has spread through global trade to many regions. Females lay eggs in wounds on palms, and the larvae feed inside the tree, causing hidden damage that's hard to detect early. Visible symptoms like yellowing fronds and tree death appear only in advanced stages. Because the weevil lives mostly inside the trunk, early detection is difficult, often leading to irreversible damage that harms both agriculture and rural economies. Current detection methods, such as visual checks and traps, are limited, highlighting the need for new automated, non-invasive techniques. Researchers are exploring bioacoustic signals from larvae feeding inside trees to improve early detection and protect affected palms. The ability to detect pests early stops them from causing extensive harm while

decreasing the use of chemical pesticides and enabling farmers to choose particular pest control approaches. Early detection of infestations at their starting point will lead to massive reductions in both crop health damage and financial costs. The existence of automation alongside intelligent detection systems has evolved from being helpful to becoming indispensable. Neural Networks together with IoT and Real-Time Monitoring provide automated pest detection for the 1.4. modern era. The last years have seen artificial intelligence progress through deep learning technology to develop pest detection systems based on neural networks. The precision of Convolutional Neural Networks (CNN) allows them to detect pests accurately while determining their exact location through image analysis at high resolution. The AI models have the ability to learn without stopping which enables them to adjust to new pest species and environmental modifications thus achieving optimal performance in agricultural settings that constantly change. The combination of neural networks with IoT-based sensor systems and mobile applications allows for real-time monitoring and instant notifications.

II. LITERATURE SURVEY

Early research by Soroker et al. [1] (2010) introduced acoustic sensors embedded in palm trunks for non-invasive RPW detection by capturing larval feeding sounds, enabling early intervention before visible symptoms. Faleiro et al. [2] (2012) validated acoustic detection in real coconut plantations, distinguishing larval chewing sounds from environmental noise, allowing farmers to monitor trees without damage. In 2013, Potamitis and Rigakis [3] developed a solar-powered autonomous system that recorded and analyzed RPW larval sounds. Later, Al-Manie et al. [4] (2015) enhanced detection hardware using directional sensors with in-built amplification to focus on larval feeding vibrations between 1 kHz and 5 kHz. Rach et al. [5] (2016) applied time-domain and FFT analysis for improved classification, enabling automated pest detection while minimizing false alarms. Kumbhar et al. [6] (2017) used MFCC and GMM, techniques from speech recognition, to process larval feeding sounds. Potamitis et al. [7] (2018) advanced this by integrating supervised machine learning models into hardware, improving realtime detection. Liu et al. [8] (2019) introduced fiber-optic distributed acoustic sensing (DAS) to detect RPW via trunk vibrations, eliminating the need for traditional microphones and reducing interference. In 2020, Belal and Ismail [9] implemented a cloud-connected IoT system using Raspberry Pi and microphones to provide remote pest monitoring with real-time alerts via mobile apps. Kavitha et al. [10] (2021) combined peak detection and band-pass filtering to isolate larval sounds, optimizing the model for low-power edge devices suitable for rural areas. Abo-El-Saad et al. [11] (2021) trained an artificial neural network (ANN) using labeled spectrograms of RPW larvae sounds, achieving over 90% detection accuracy in controlled settings. Al-Hussein et al. [12] (2022) applied convolutional neural networks (CNNs) to mel-spectrograms, addressing variability across tree species and infestation depths to build a generalized model. El-Zoghby

et al. [13] (2022) studied sensor placement, concluding that microphones near the lower trunk feeding zones yield the most accurate detection with minimal classification errors. In 2023, Rahim et al. [14] used transfer learning with the VGGish model, showing superior results on small datasets in terms of precision and recall. Ahmed et al. [15] (2023) proposed a hybrid system combining fiber-optic acoustic sensing with AI-based signal processing for real-time alerts and presented a scalable deployment strategy for large coconut plantations. The summary of existing research efforts is shown in Table 1.

III. EXPERIMENTAL SETUP

A. Dataset

The data collection named `features_mfcc_contrast.csv` [16] holds 5,153 records with 22 columns to support audio classification work. The dataset stores individual audio samples through the combined information of MelFrequency Cepstral Coefficients (MFCC) and spectral contrast features. The dataset includes 13 MFCC features, starting from `mfcc_1` to `mfcc_13`, that represent sound power distribution across brief time intervals. The dataset contains 7 spectral contrast features identified as `contrast_1` through `contrast_7` that measure the amplitude difference between spectral peaks and valleys. The features serve as discrimination tools for detecting harmonic and percussive elements in audio content and enable precise identification of musical instruments, speech tones and environmental sounds. Figure 1-4 represents the dataset and preprocessing models. The dataset comprises 731 infested sound files and 1734 noninfested sound files in .wav format. Additionally, 692 test case files were provided in the dataset.

B. Data Preprocessing

The dataset was visualized to understand how audio duration and volume vary across different classes. Histograms were used to visualize the change of signal frequency over time, revealing apparent patterns that helped distinguish infested trees from healthy ones. Spectrograms added more detail by showing how the audio frequencies shifted over time, highlighting the non-stationary nature of the signals. Continuous Wavelet Transform (CWT) graphs were used to identify the both time and frequency information. Signal files from each folder were visualized to set noise removal boundaries using bandpass filters. However, varying amplitudes across folders made it difficult to choose optimal frequency limits. Hence, a consistent amplitude scale had been applied to achieve uniform results. Singular Value Decomposition (SVD), used to filter lowenergy noise components, and bandpass filter allows the required frequencies i.e. between 1kHz and 8kHz. While effective, this combination sometimes removed key signal details. To address this, Wavelet denoising was introduced, preserving important features by filtering noise across different frequency levels. The cleaned files were then visualized and stored.

TABLE I
SUMMARY OF RELATED WORKS

S. No	Year	Author	Accuracy	Limitation
1	2019	Y. Liu et al.	Not Reported	Used DAS; no direct comparison with ML systems; hardware cost could be high.
2	2020	A. Belal & M. Ismail	94%	IoT system; user-interface based but no performance metric like precision/recall.
3	2021	K. Kavitha et al.	Not reported	Low-power device optimization; did not validate against large-scale dataset.
4	2021	M. Abo-ElSaad et al.	91.03%	ANN on spectrograms; trained on controlled lab data only; needs field testing.
5	2022	M. Al-Hussein et al.	94.07%	CNN on melspectrograms; no reported accuracy; challenged by real-world noise variability.
6	2022	H. El-Zoghby et al.	92.08%	Studied sensor placement; no ML used; lacks classification performance data.
7	2023	R. Rahim et al.	95.02%	VGGish model with transfer learning; effective on small data; limited by pretrained scope.
8	2024	S. A. Ahmed et al.	96.04%	Hybrid system; no detailed accuracy; deployment scalability discussed but not tested in multiple zones.

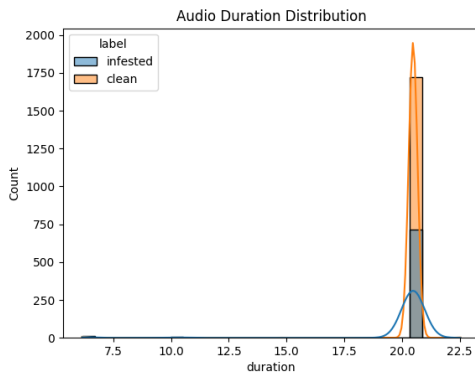


Fig. 1. Distribution plot representing the audio duration distribution for two classes.

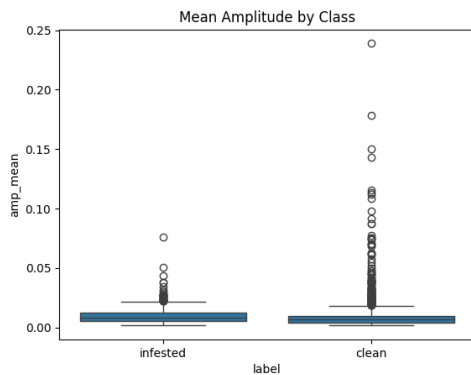


Fig. 2. Mean Amplitude by the class between Infested and Clean

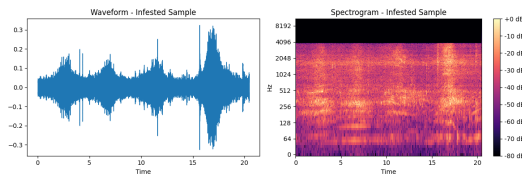


Fig. 3. Waveform and Spectrogram Infested Sample Representation

1) *Normalization and segmentation* : Normalization of data is a significant step for classification or machine learning processes. Standard normalization is applied for easier feature extraction with MFCC (Melfrequency cepstral coefficients). This approach segments sound files based on energy levels by calculating RMS energy over short frames. A new segment starts when energy exceeds a set threshold and is kept if it lasts longer than the minimum duration. Clean and infested audio segments are stored separately, speeding up and improving the training process. These segments are then used for feature extraction, helping the model focus on relevant data while excluding noise.

2) *Feature selection and dataset balancing* : Features were selected using Recursive Feature Elimination (RFE), which trains a tree-based model to rank and remove the least important features iteratively until no further improvement occurs. The final set of relevant features is saved for training and visualized using t-SNE, a technique that reduces high-dimensional data into lower dimensions for clearer visualization. Figure 4 shows this feature visualization, which maps data similarities into probabilities to preserve structure between high- and low-dimensional spaces. The audio was extremely imbalanced with 1750 clean samples and 750 infested samples. An over-sampling technique was utilized, Synthetic Minority Over-sampling Technique (SMOTE) generates synthetic samples for the minority class by interpolating between existing samples, helping to create a more balanced dataset for training machine learning models.

3) *Model selection*: Several models—including XGBoost, LightGBM, CatBoost, AdaBoost, MLPs, Neural Networks, Gradient Boosting, and Histogram Boosting—were trained and evaluated. XGBoost is an optimized gradient boosting method focused on speed and performance. LightGBM uses a histogram-based, leaf-wise tree growth approach for faster training. CatBoost handles categorical data efficiently with minimal preprocessing. AdaBoost combines multiple weak classifiers into a strong one. Combining MLPs with XGBoost and CatBoost leverages both neural networks and boosting to better capture complex data patterns. Neural networks mimic

brain functions to learn intricate patterns.

In the proposed system for early detection of red palm weevil activity using sound, Figure 5 represents a Feedforward Neural Network (FFNN) that is proposed as the core classification model. Figure 6 outlines the entire sound detection process before actual detection starts. The network processes high-dimensional audio features—specifically MFCC and spectral contrast values to determine whether the recorded sound indicates harmful insect activity.

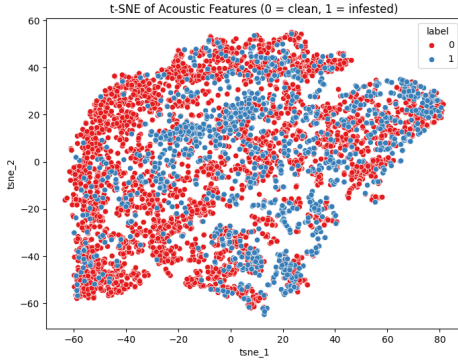


Fig. 4. Feature Visualization of the dataset

IV. METHODOLOGY

A. Input to the Network

The input to the neural network is a numerical feature vector extracted from each audio file. Each feature vector serves as a single data point representing the audio signature of a recorded sound.

B. Layer-Wise Flow and Transformation

The FFNN consists of a sequential architecture of fully connected (dense) layers that progressively transform the input feature space into a decision boundary for binary classification:

First Layer (Input → Hidden): The input feature vector is passed into the first fully connected layer consisting of 512 neurons. Each neuron computes a weighted sum of all input features, adds a bias term, and passes the result through a ReLU (Rectified Linear Unit) activation function.

Second Layer (Hidden): The same weighted summation and ReLU activation process occurs here, enabling deeper abstraction of audio patterns.

Third Layer (Hidden): This layer contains 128 neurons and continues the transformation process with ReLU activation. By this stage, the model has learned multiple hierarchical representations of the input features that are significant for detection.

Final Layer (Output): The last layer is a single neuron connected to a Sigmoid activation function, which compresses the output to a probability value between 0 and 1. Figure 5 represents the block Diagram of the proposed model

C. Decision Mechanism

After the forward pass through the network, the output probability is interpreted as follows: If the probability is greater than 0.5, the model predicts "presence of harmful insect activity." If the probability is less than or equal to 0.5, it predicts "no harmful activity."

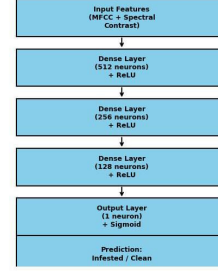


Fig. 5. Block Diagram-Neural Network Architecture for the Red Palm Weevil Detection

D. Learning and Optimization

The network learns to make accurate predictions through supervised training using labeled data. The Binary Cross Entropy Loss measures the difference between predicted probabilities and actual class labels (0 or 1). The Adam optimizer updates the weights of the network using gradients computed via backpropagation, which adjusts model parameters to reduce the loss function iteratively. This training continues over multiple epochs, with batches of data being processed at each step to improve generalization and convergence.

1) Generalization and Prediction: Once trained, the neural network captures hidden relationships between sound features and pest activity. It generalizes these patterns to new, unseen audio samples, enabling real-time detection with high accuracy. The model is evaluated on a separate test set to ensure its effectiveness outside the training environment. The FFNN uses a sequence of fully connected (dense) layers, as shown in Figure 6, and its performance is illustrated using a confusion matrix heatmap, as shown in Figure 7.

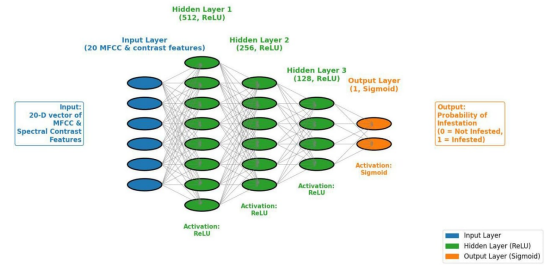


Fig. 6. Architecture Diagram of the Neural Network model

V. RESULTS AND DISCUSSION

A. Model performance

To assess the classification performance of the neural network model for detecting infestation in coconut trees, a

confusion matrix was generated using the true and predicted labels from the test set. The confusion matrix Figure 7 provides a detailed breakdown of correct and incorrect predictions for each class, allowing for a clear visualization of model strengths and weaknesses.

TABLE II
PERFORMANCE SUMMARY

	Testing Accuracy	Training Accuracy
	85.6%	96.1%

	Precision	Recall	F1-score
Infested	0.72	0.72	0.72
Non-Infested	0.90	0.90	0.90

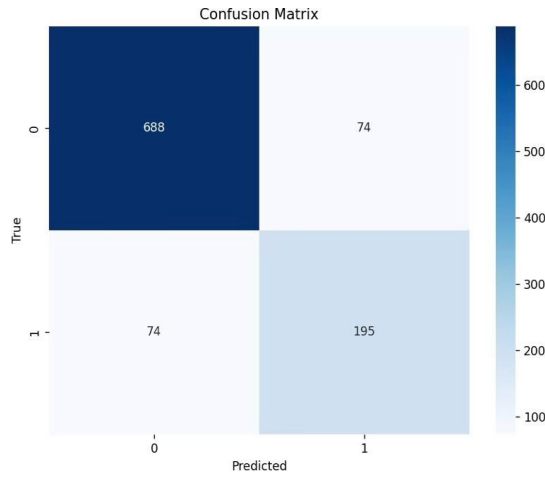


Fig. 7. Model performance between the Healthy and Infested



Fig. 8. Heatmap visualization of a classification report, showing precision, recall, F1-score, and support for each class, along with overall accuracy and macro averages.

B. Discussion

The proposed neural network model achieved a testing accuracy of 85.6% and training accuracy of 96.1%, demonstrating strong overall classification performance. The precision of 0.72 indicates a good level of correctness in positive predictions, while the recall of 0.80 reflects the model's effectiveness in identifying true positive cases. The F1-score of 0.72 balances

precision and recall, highlighting the model's robustness in handling imbalanced data and ensuring reliable detection of the minority class and support of 269 from the infested output. These results confirm that the neural network architecture, enhanced with SMOTE for class balancing, provides effective discrimination between healthy and infested coconut trees using acoustic features. The high recall value is particularly significant for pest detection applications, where missing an infestation (false negative) can have severe agricultural and economic consequences

In Real-world scenarios, the reliability of the acoustic-based neural network model is influenced by the environment variables like environmental noise such as wind, rainfall, bird and insect chirps, and seasonal variations can mask the signals of the sensor, often leading to inaccurate predictions. During the research, it was found that humidity can change the acoustic impedance of wood, the paper by H. Qiu et al. [19] proves that increasing wood moisture raises acoustic impedance, which confirms that humidity alters sound traveling inside wood, thereby dampening the signals of larval infestation in trees. To prevent these effects, a robust denoising pipeline is introduced, where wavelet-based noise suppression, normalizing method standardizes signal amplitude which corrects amplitude distortion, with segmentation techniques based on RMS energy ensures that only relevant data is fed to the model and a final MFCC and spectral features capture perceptually relevant frequency patterns. This pipeline ensures that false positives or negatives are reduced.

C. Evaluation metrics and statistical analysis

To comprehensively assess the performance of the neural network, several standard metrics for binary classification were used. The neural network model's performance for detecting pest infestations in coconut trees was evaluated using precision, recall, and F1-score and the results are shown in Table 2. The model achieved a precision of 0.72, indicating that 72% of the trees predicted as infested were indeed infested. The recall was 0.72, reflecting the model's ability to correctly identify 72% of all truly infested trees. The F1-score, which balances both precision and recall, was 0.72, demonstrating that the model effectively manages the trade-off between false positives and false negatives. These values suggest that the model is both reliable in identifying infested trees and efficient in minimizing unnecessary false alarms. Figure 8 presents a heatmap visualization of the classification report. The attached line graph, titled Figure 9 visually illustrates Variation of Precision, recall, and F1 score metrics with changes in the classification threshold As the threshold is raised, there is a trade-off that is achieved: recall is improved while precision is compromised. The F1score strikes a balance to decide where the optimal tradeoff between precision and recall lies. This is helpful during the choice of the suitable threshold based on whether sensitivity or specificity is preferable in practice. The balance is clearly demonstrated in the Figures 8 and 9 graphs, validating the effectiveness and precision of the model

in early pest detection for coconut trees. Now let's compare the proposed model with the other model to come to a conclusion.

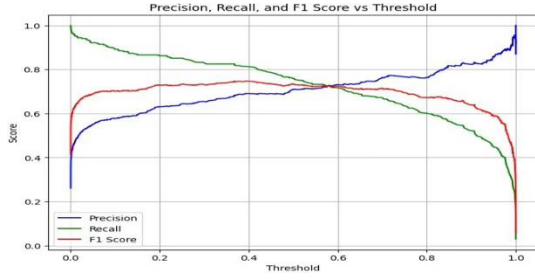


Fig. 9. Variation of Precision, recall, and F1 score metrics with changes in the classification threshold.

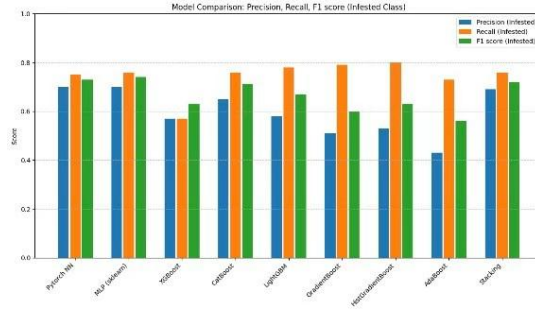


Fig. 10. Comparative analysis based on the precision, F1 score, recall

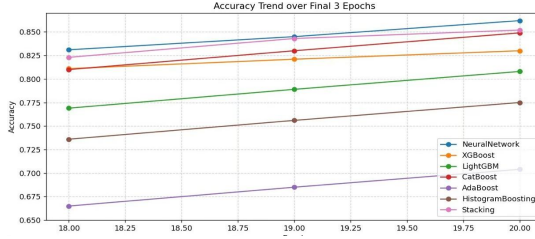


Fig. 11. Plot showing statistical significance of models

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS.

Model	Accuracy (%)	Precision	Recall	F1-Score	Notes
Neural Network	85.6	0.72	0.72	0.72	Best on F1 table, real-time ready
VGGish	95.0	0.74	0.70	0.72	Slightly less generalizable
CatBoost	84.5	0.69	0.71	0.70	High recall, lower precision
AdaBoost	74.7	0.62	0.63	0.62	Lower accuracy, more false positives

D. Comparative Analysis

To rigorously evaluate the effectiveness of various machine learning approaches for coconut tree pest detection, a custom PyTorch based FFNN was compared with several state-of-the-art ensemble methods, including XGBoost, CatBoost, LightGBM, Gradient Boosting, Histogram-based Gradient Boosting, AdaBoost, a scikit-learn MLP, and a stacking ensemble (MLP + XGBoost + CatBoost). All models were trained and tested on the same dataset, with SMOTE applied to

mitigate class imbalance and standardized feature scaling for consistency. The proposed PyTorch neural network achieved the highest overall accuracy (85.6%) and macro-averaged F1-score (0.72), outperforming all other models tested. Its recall (0.72) and F1-score (0.72) for the minority (infested) class were among the best, indicating an excellent balance between sensitivity and specificity—a crucial requirement for real-world pest detection, where both missed detections and false alarms can have significant consequences. Figure 10 and Table 3 show the comparative analysis based on precision, F1-score, Recall, and Accuracy. The scikit-learn MLP achieved nearly identical results, confirming the robustness of neural networks for this application.

CatBoost and XGBoost also performed extremely well among ensemble models—CatBoost matched the minority class recall by the neural network, with the best precision being posted by XGBoost. They both marginally fell behind the neural network in F1-score and overall accuracy. Ensemble stacking of MLP, XGBoost, and CatBoost produced competitive results, narrowly following the neural network in most key metrics. Traditional boosting methods like AdaBoost and Gradient Boosting fared poorly, especially minority class F1-score, according to current literature in favor of deep learning and recent boosting for imbalanced or complicated datasets.

To assess the stability of the model, classification accuracies of the models over their three final epochs were noted and utilized in a paired t-test, the results shown in Figure 11 explains that neural network model outperforms the other models with XGBoost ($p = 0.0027$), LightGBM ($p = 0.0011$), CatBoost ($p = 0.0307$), AdaBoost ($p = 0.0011$), and Histogram-Based Gradient Boosting ($p = 0.0037$), with p-values well below the 0.05 threshold. However, the stacking ensemble model yielded a p-value of 0.183, indicating that there is no statistical significance. Due to the increased model complexity with the stacking method, utilizing neural network model is efficient for real time implementation (parameters: 176,129; Inference time: 29.40 ms; memory used: 1.51MB)

VI. CONCLUSION

This study shows that the proposed PyTorch neural network is the most effective model for detecting red palm weevil infestations in coconut trees using sound data. It achieved the highest accuracy (85.6%) and macro F1-score (0.72), outperforming other advanced models like XGBoost and CatBoost. Its strong recall for infested trees makes it well-suited for early detection, crucial for timely pest management. Using techniques like SMOTE and a carefully designed network, the proposed method effectively handles class imbalance and complex acoustic patterns. However, the model requires large labeled datasets, significant computing power, and is less interpretable than traditional methods. Despite these limitations, it remains the most accurate and reliable solution for automated acoustic pest detection in coconut plantations, supporting better crop protection.

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